Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 05)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 1 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 1 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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0 2 . 1 Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

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0 2 . 2 Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

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0 2 . 3 Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

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0 2 . 4 Explain what is meant by **synergism** and give an example.

[3 marks]

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0 3 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☐

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

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0 3 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 3 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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0 4 A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 4 . 1 Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 4 . 2 Give **three** limitations of kick sampling.

[3 marks]

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0 4 . 3 Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 4 . 4 Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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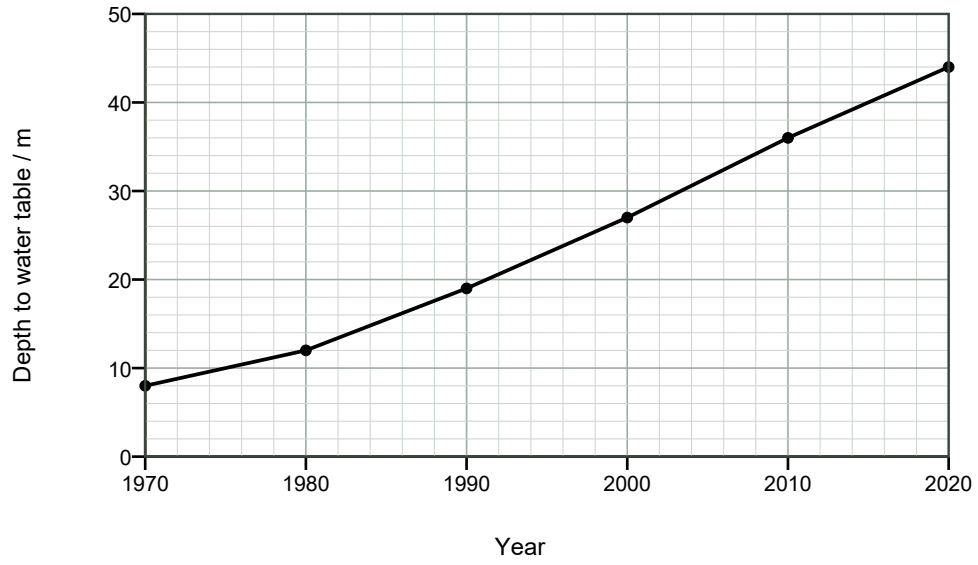
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0 5 Figure 1 shows the depth of the water table below the ground surface in an aquifer beneath a coastal farming region between 1970 and 2020.

Figure 1



0 5 . 1 Calculate the mean rate at which the water table fell between 1970 and 2020.

[1 mark]

Answer ☐ m/yr

0 5 . 2 Explain why the water table is falling.

[2 marks]

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0 5 . 3 Explain the features a rock must have to form an aquifer.

[3 marks]

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0 5 . 4 Describe **four** consequences of over-exploiting this coastal aquifer.

[4 marks]

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0 6 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

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0 6 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties .....

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Properties causing ozone depletion .....

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0 6 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

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0 6 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 6 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 6 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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0 7 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 7 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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0 7 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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0 8 Table 1 shows the annual mean concentrations of four pollutants measured at a city-centre monitoring station, with the legal limit for each.

Table 1

Pollutant	Annual mean / μg m^{-3}	Legal limit / μg m^{-3}
Nitrogen dioxide	52	40
PM10 particulates	28	40
PM2.5 particulates	19	20
Sulfur dioxide	8	125

0 8 . 1 Calculate by what percentage the annual mean nitrogen dioxide concentration exceeds the legal limit.

[2 marks]

Answer ..... %

0 8 . 2 Suggest why the sulfur dioxide concentration is now far below its limit while nitrogen dioxide is not.

[3 marks]

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0 8 . 3 Explain how each of the following reduces emissions.

[3 marks]

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Electrostatic precipitator .....

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Catalytic converter .....

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Low-NOx burner .....

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0 8 . 4 Explain why a reduction in the city's NOx emissions would also reduce summer ground-level ozone.

[3 marks]

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0 8 . 5 The council also plans to monitor air quality using lichens. Explain the advantages and the limitations of using lichens as indicator species.

[4 marks]

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0 9 A low-lying coastal town in eastern England is at risk of flooding from rising sea level and more intense rainfall.

0 9 . 1 Describe **three** ways the town could adapt to the increased flood risk.

[3 marks]

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0 9 . 2 Explain the circumstances in which managed retreat may be chosen instead of coastal defences.

[2 marks]

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0 9 . 3 Describe **two** impacts of higher temperatures on transport infrastructure.

[2 marks]

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0 9 . 4 Explain how carbon capture and storage (CCS) could reduce CO_2 emissions from a fossil fuel power station, and give **one** limitation.

[3 marks]

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1 0 . 1 Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

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Extra space

1 0 . 2 State what is meant by the **residence time** of a substance in a reservoir.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS